

Hence, $X_{\pm 3} = \emptyset$, and (4.2) follows. Furthermore, we see that for every vector $\gamma = (\gamma_1, \gamma_2, \gamma_3) \in X$, γ_1, γ_2 and γ_3 are minimal representatives in $m\alpha_9 + A_9$, $2m\alpha_9 + A_9$ and $\alpha_1 + A_1$, respectively. By Lemma 3.1 again, we have (4.3) and (4.4). $\square$

By direct calculation, we have the following.

Lemma 4.3. *Let $i, j, k \in [10]$ with $j \neq k$, and $i', j', k' \in [10]$ with $j' \neq k'$. Then,*

$$(v_\varepsilon(i, j, k), v_\varepsilon(i', j', k')) = |\{i\} \cap \{i'\}| + |\{j, k\} \cap \{j', k'\}| \quad (\varepsilon = \pm 1). \quad (4.5)$$

Let $I, J \in \binom{[10]}{5}$. Then,

$$(v_1(i, j, k), v_5(I)) = |\{i\} \cap I|, \quad (4.6)$$

$$(v_5(I), v_5(J)) = |I \cap J| - 2. \quad (4.7)$$

Definition 4.4. Let $\mathcal{X}$, $\mathcal{X}_1$ and $\mathcal{X}_5$ be the sets of maximum affine equiangular sets in X , X_1 , and X_5 , respectively, with respect to r .

Our aim is to investigate maximum affine equiangular sets ω in X with norm 3 with respect to r up to switching equivalence, that is, replacing some of the vectors $u \in \omega$ by $r - u$. From the following lemma, we may restrict our attention to affine equiangular sets in X_1 and ones in X_5 .

Lemma 4.5. *We have $\mathcal{X}/\sim_{\text{sw}} = \{[\psi \cup \phi] : [\psi] \in \mathcal{X}_5/\sim_{\text{sw}}, \phi \in \mathcal{X}_1\}$. In particular, $|\mathcal{X}/\sim_{\text{sw}}| = |\mathcal{X}_5/\sim_{\text{sw}}| |\mathcal{X}_1|$.*

Proof. Let $\mathcal{X}_{\pm 1}$ be the set of maximum affine equiangular sets in $X_1 \cup X_{-1}$. By (4.6), we have $(u, v) \in \{0, 1\}$ for any $u \in X_1 \cup X_{-1}$ and $v \in X_5$. Hence,

$$\mathcal{X} = \{\psi \cup \phi : \psi \in \mathcal{X}_5, \phi \in \mathcal{X}_{\pm 1}\}.$$

Since $X_{-1} = \{r - u : u \in X_1\}$, we see that for each $\phi \in \mathcal{X}_{\pm 1}$, there exists (a unique) $\phi' \in \mathcal{X}_1$ with $\phi' \sim_{\text{sw}} \phi$. Noting that X_5 and $X_{-1} \cup X_1$ are closed under switching equivalence $\sim_{\text{sw}}$, we have

$$\begin{aligned} \mathcal{X}/\sim_{\text{sw}} &= \{[\psi \cup \phi] : [\psi] \in \mathcal{X}_5/\sim_{\text{sw}}, [\phi] \in \mathcal{X}_{\pm 1}/\sim_{\text{sw}}\} \\ &= \{[\psi \cup \phi] : [\psi] \in \mathcal{X}_5/\sim_{\text{sw}}, \phi \in \mathcal{X}_1\}. \end{aligned}$$

Next, we verify $|\mathcal{X}/\sim_{\text{sw}}| = |\mathcal{X}_5/\sim_{\text{sw}}| |\mathcal{X}_1|$. Let $\psi, \psi' \in \mathcal{X}_5$ and $\phi, \phi' \in \mathcal{X}_1$ be such that $[\psi \cup \phi] = [\psi' \cup \phi']$. Then, it suffices to show $[\psi] = [\psi']$ and $\phi = \phi'$. Indeed, we obtain $[\psi] = [\psi']$ and $[\phi] = [\phi']$. Moreover, as discussed earlier, since an element of $\mathcal{X}_1$ uniquely exists as a representative of $[\phi]$, we conclude $\phi = \phi'$. $\square$

5. MAXIMUM AFFINE EQUIANGULAR SETS IN X_5

In this section, we determine a unique maximum affine equiangular set in X_5 up to $\text{Aut}(\Lambda)_r$ with a computer.

Lemma 5.1. *Every affine equiangular set in X_5 with respect to r is of cardinality at most 12.*

Proof. Recall that the vectors in X_5 are

$$\left(\frac{1}{2}\mathbf{1} - \sum_{i \in I} e_i, 0, \alpha_1 \right) \quad \text{for } I \in \binom{[10]}{5}.$$

Hence, the dimension of the linear space generated by X_5 together with r is at most $\dim(\mathbb{R}^{10} \perp 1) + 1 = 10$. Thus, an arbitrary affine equiangular set in X_5 with respect to r induces a set of equiangular lines with common angle $\arccos(1/5)$ in dimension 9. The cardinality $N_{1/5}(9)$ of a maximum set of equiangular lines with common angle $\arccos(1/5)$ in dimension 9 equals 12 [11,